

High-Level Design

A compliant, observable outbound collections assistant

Maya is designed to request a resolution without pressure: authenticate first, disclose only what the backend permits, and leave a clear, reviewable disposition.

SUBMISSION SCOPE

- Working React/Vite dashboard
- Express state-enforced webhook
- Five strict Vapi tools
- Synthetic in-memory account

Approved Clara reference is included locally.
Live Vapi/TTS remains a manual setup and real-call verification step.

Design principles

- 01 Backend authority over authentication
- 02 Zero debt disclosure before verification
- 03 Narrow tools with structured failures
- 04 Honest status reporting

Key target

- < 1.2 seconds target conversational latency
- ≈ 1.1 seconds budget; not a measured guarantee

STATUS BOUNDARY

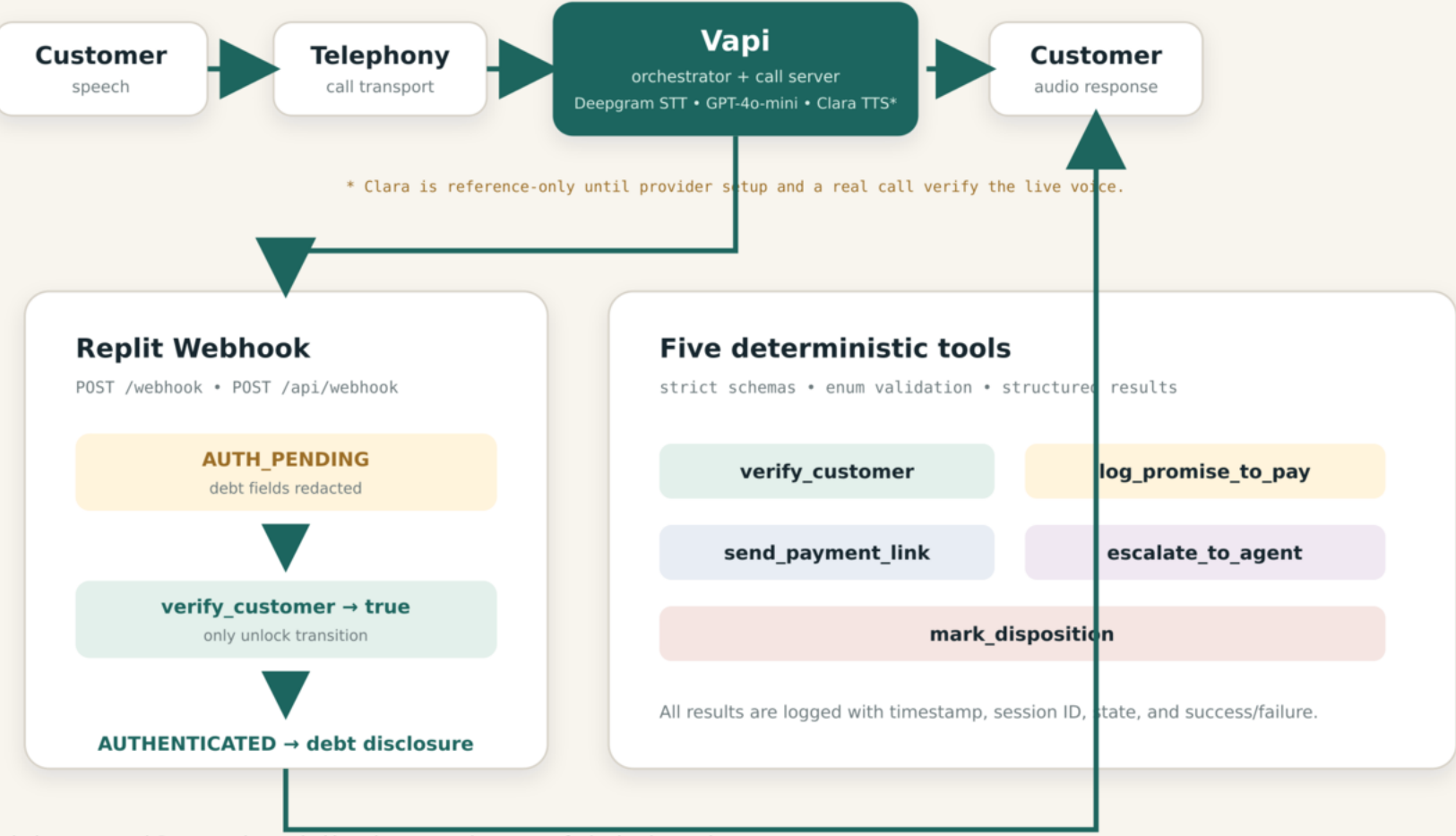
Verified locally: API contract, guardrail, tools, scenarios, dashboard, and tests.
Requires manual Vapi setup: assistant, HTTPS webhook, tools, and provider Clara voice.
Not yet verified: real Vapi call, live TTS, and hearing Clara through the provider.

Architecture

s in Replit.

Maya / Kapture Finance — System Architecture

BACKEND-ENFORCED AUTHENTICATION • VAPI-READY CONTRACT • SAFE ACCOUNT PROJECTION



Design Details

Runtime design

Pipeline

- 1 Calling-hours gate: 08:00–19:00 customer's local time
- 2 Telephony → Vapi → Deepgram Nova-2 → GPT-4o-mini
- 3 Intent/entity extraction selects one strict backend tool
- 4 Webhook validates, enforces state, and returns matching toolCallId
- 5 Vapi speaks only confirmed results using provider-configured Clara

State machine

INIT → AUTH_PENDING → AUTHENTICATED → NEGOTIATION → ACTION → CALL_ENDED

Safe exits: wrong person, verification failure, DNC, no response, escalation.

Authentication and data safety

Before AUTHENTICATED: account ID and customer name only; loan, EMI, amount, debt, and days past due are null. Only verify_customer with verified:true unlocks projection. Verification values stay in the backend mock record and are never logged or returned.

Guardrails

Maya does not threaten, shame, argue, invent fees/waivers, or fabricate payment confirmation. A DNC request ends negotiation immediately. Disputes, hardship, callbacks, and hostility escalate.

Disposition

PTP_AGREED, ALREADY_PAID, DISPUTED, HARDSHIP_ESCALATED, WRONG_PERSON, DO_NOT_CALL, CALLBACK_REQUEST, VERIFICATION_FAILED, NO_RESPONSE, HOSTILE_CALLER

Validation & Handoff

Operations, testing, and handoff

Observability

Dashboard/API metrics: total calls, PTP rate/count, containment, escalation, verification-failure, already-paid, dispute, hardship, DNC, no-response, average tool latency, average call duration, disposition completion, tool success.

Testing

TC-001 through TC-015 are documented in tests/test_cases.json.
tests/run_tests.mjs executes the API security, scenario, tool-failure, invalid-account, and malformed-webhook checks against the running API workflow.

MANUAL VAPI SETUP

- 1 Create Maya — Kapture Finance Collections
- 2 Configure GPT-4o-mini, temperature 0.1, Deepgram Nova-2
- 3 Configure approved Clara using provider-supported voice/custom voice flow
- 4 Paste system_prompt.txt, register tool_definitions.json, set https://YOUR_PUBLIC_HOST/webhook
- 5 Place PTP, already-paid, and DNC calls during 08:00–19:00 local time

Known limitations

In-memory data resets on restart. Telephony, Vapi, payments, identity verification, callback scheduling, and provider TTS are intentionally not connected in this take-home.

Final status

VERIFIED: local backend, dashboard, contract, guardrail, scenarios, and API checks.
REQUIRES MANUAL VAPI SETUP: provider account, HTTPS URL, tools, and Clara configuration.
NOT YET VERIFIED: real Vapi call, live STT/TTS, and heard Clara voice.